

Jonas Conneryd

Department of Computer Science
Lund University, Sweden

Date of birth: July 31st, 1997
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Education

Lund University

Ph.D. in Theoretical Computer Science
Advisor: Prof. Tatyana Turova

Lund, Sweden
2026 (expected)

Lic. Eng. in Computer Science

Thesis defended May 2025

Thesis: *On the Average-Case Proof Complexity of Graph Coloring*

Opponent: Prof. Nutan Limaye, ITU Copenhagen

KTH Royal Institute of Technology

Stockholm, Sweden

M.Sc. in Mathematics (joint with Stockholm University)

June 2021

Thesis: *Geometric Bounds for Steklov Eigenvalues on Graphs* (awarded Mittag-Leffler Prize)

B.Sc. in Engineering Physics

June 2019

Thesis: *Explicit Symplectic Integrators for Non-Separable Hamiltonians in Molecular Dynamics*

Research Interests

Computational complexity theory, particularly proof complexity with an algebraic flavor and the complexity of (random) constraint satisfaction problems.

Publications

- [1] Jonas Conneryd, Yassine Ghannane, and Shuo Pang. *Lower Bounds for CSP Hierarchies Through Ideal Reduction*. SODA 2026.
- [2] Jonas Conneryd, Kilian Risse, and Dmitry Sokolov. *Graph Coloring Is Hard on Average for Polynomial Calculus over Roots of Unity*. To be incorporated into a larger project also involving Yassine Ghannane, Jakob Nordström, and Shuo Pang. 2025.
- [3] Jonas Conneryd, Susanna F. de Rezende, Jakob Nordström, Shuo Pang, and Kilian Risse. *Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz*. FOCS 2023.
- [4] Jonas Conneryd. *Geometric Bounds for Steklov Eigenvalues on Graphs*. M.Sc. thesis. Stockholm University, 2021.

Talks

ACM-SIAM Symposium on Discrete Algorithms (SODA)

Lower Bounds for CSP Hierarchies Through Ideal Reduction

Vancouver, BC, Canada
Winter 2026

KTH Royal Institute of Technology

Lower Bounds for CSP Hierarchies Through Ideal Reduction

Stockholm, Sweden
Fall 2025

Oxford Proof Complexity Workshop

Lower Bounds for CSP Hierarchies Through Ideal Reduction

Oxford, United Kingdom
Summer 2025

Proof Complexity and Beyond, Oberwolfach Workshop 2413

Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz

Oberwolfach, Germany
Spring 2024

Algorithmic Research Cooperation around Øresund (ARCO)

Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz

Odense, Denmark
Fall 2023

IEEE Symposium on Foundations of Computer Science (FOCS)

Graph Colouring Is Hard on Average for Polynomial Calculus and Nullstellensatz

Santa Cruz, CA, USA
Fall 2023

Research Visits and Workshops

KTH Royal Institute of Technology

Month-long visit to the theory group.

Stockholm, Sweden
Fall 2025

University of Oxford

Visiting researcher for the workshop *Proof Complexity*.

Oxford, United Kingdom
Summers 2024, 2025

Centre International de Rencontres Mathématiques

Visiting researcher for the *Complexity as a Kaleidoscope* research school.

Marseille, France
Spring 2025

DIMACS, Rutgers University

Visiting researcher for the *Frontiers in Complexity Workshop*.

New Brunswick, NJ, USA
Summer 2024

McGill University

Week-long visit to the group of Prof. Robert Robere.

Montréal, Canada
Summer 2024

Mathematisches Forschungsinstitut Oberwolfach

Visiting researcher for the workshop *Proof Complexity and Beyond*.

Oberwolfach, Germany
Spring 2024

Chalmers University of Technology

Visiting researcher for the *Workshop on Algebra and Computation*.

Gothenburg, Sweden
Summer 2023

Simons Institute for the Theory of Computing at UC Berkeley
2-month visiting graduate student for the program *Satisfiability: Extended Reunion*.
Schloss Dagstuhl
Visiting researcher for the workshop *Satisfiability: Theory, Practice and Beyond*.

Berkeley, CA, USA
Spring 2023
Dagstuhl, Germany
Fall 2022

Honors and Scholarships

- *Oberwolfach Leibniz Graduate Student*; Mathematisches Forschungsinstitut Oberwolfach, 2024
- Accepted as *WASP* affiliated student, 2021
- *Mittag-Leffler Prize for outstanding M.Sc. theses in mathematics*; Stockholm University, 2021
- *Ingenjör Ernst Johnson Scholarship for outstanding academic achievements*; KTH, 2020, 2021
- *Henrik Göransson Sandviken Scholarship for outstanding academic achievements*; KTH, 2018
- *University Merit Scholarship for outstanding academic achievements*; KTH, 2018, 2019, 2020, 2021

Service

Reviewed papers for the following venues:

- Conferences: FOCS '22, CCC '22, CCC '23, SAT '23, SAT '24, ICALP '24, AAI '23, AAI '24, SOSA '25, STOC '25
- Journals: Transactions on Computational Logic, Theoretical Computer Science, SIAM Journal on Discrete Mathematics

Teaching Experience

Lund University

- EDAA40/75 Discrete Structures in Computer Science, 2022, 2024, 2025
- EDAN55 Advanced Algorithms, 2023, 2024
- EDAN01 Constraint Programming, 2022, 2023

KTH Royal Institute of Technology

- SF1661 Perspectives on Mathematics, 2018
- SF1624 Algebra and Geometry, 2017

Work Experience

AP3 Third National Swedish Pension Fund
ILS Intern
Quantitative analysis of insurance-linked securities (ILS).

Stockholm, Sweden
2019-2021

Technical Skills

Programming: Python, Go, $\text{\LaTeX}$, Julia, MATLAB
Software: RMS Miu